Cross-Benchmark Failure Analysis

English vs Bangla LLM agents on OfficeBench, τ²-bench, WorkBench and TheAgentCompany

Results and analysis · where and why models fail · root causes · trajectory-level comparison · evidence and recommendations

	OfficeBench	τ²-bench	WorkBench	TheAgentCompany
Interaction	Single-turn office tasks in a container (≤ 20 steps)	Customer-service dialogue with an LLM user simulator	Single-turn function calling against a sandbox DB	Long-horizon company tasks (OpenHands) with LLM coworkers
Models	gemma4:31b, gpt-oss:20b	gemma4:31b, gpt-oss:20b	gemma4:31b, gpt-oss:20b	gemma4:31b
Paired tasks	300 subtasks	218 (gemma), 149 (gpt-oss), +219 teammate set	90 (15 × 6 domains)	23 task-runs (21 tasks)
Scoring	Benchmark checks on final files (3 scorer defects fixed)	DB state, actions, info, device state (reward ≥ 1)	Final DB = ground truth and every call executes	Weighted checkpoints; pass = all points
Failures analysed by hand	42 + 25 EN-pass/BN-fail	40 + 26 EN-pass/BN-fail	All 112 failed runs (22 EN-pass/BN-fail)	8 selected cases

Source material. This report synthesises four failure-analysis reports generated on 2026-10-10 with the same pipeline (failure_analysis_pipeline.md): OFFICEBENCH_, TAU2_, WORKBENCH_ and TAC_FAILURE_ANALYSIS.pdf. Every number is transcribed from those reports; nothing was re-run for this synthesis. Where a source report flags a number as heuristic, provisional or single-annotator, that caveat is carried over.

Evidence tags. **[OBSERVED]** a count, rate, test or trace excerpt reported in a source. **[INTERPRETATION]** a reading of what observed facts mean. **[HYPOTHESIS]** a plausible mechanism not yet confirmed; a follow-up experiment is proposed for each.

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Executive summary

Key findings

- [OBSERVED] **Language effects are model- and benchmark-dependent.** gemma4:31b is worse in Bangla on OfficeBench (50.3% → 42.3%, p = 0.003) and τ²-bench (49.5% → 38.1%, p = 0.001), not on WorkBench (81.1% → 82.2%). gpt-oss:20b is *better* in Bangla on OfficeBench (28.7% → 35.3%, p = 0.022) and unchanged on τ²-bench and WorkBench. TAC (23 task-runs) shows 39.1% → 26.1%, not significant (p = 0.25).
- [OBSERVED] **Most Bangla-only failures are not Bangla agent failures.** Of 163 hand-analysed EN-pass/BN-fail cases, only 24 (14.7%) are coded genuine Bangla agent failures (C1). The rest are the user simulator (35), checker or task ambiguity (22), translation (9), tools/environment (8), general weaknesses that also occur in English (45) and unclear cases (20).
- [OBSERVED] **The dominant confound differs by benchmark design:** English-only answer checkers on OfficeBench (15 of 42 gemma cases), the Bangla user simulator on τ²-bench (35 of 66 cases), general model weakness on WorkBench (17 of 22), run-to-run noise and environment walls on TAC.
- [OBSERVED] **Removing benchmark-side confounds shrinks every significant gap toward zero:** OfficeBench gemma +8.0 → +1.9 pp (CI [−1.9, 5.6]); τ² gemma +11.5 → +5.8 pp ([−1.9, 12.8]); OfficeBench gpt-oss −6.7 → −3.3 pp ([−7.7, 1.5]); TAC +13.0 → 0.0 pp.
- [OBSERVED] **Two genuine Bangla mechanisms recur across models and benchmarks:** (1) entity grounding from a script without letter case: names written in Bangla script are lower-cased for case-sensitive lookups (OfficeBench: 3 gemma + 6 gpt-oss cases); (2) re-rendering English values from a Bangla prompt: "Trip" for travelling, "curd" for yogurt, "08:00 AM" for 8:00, a rolling 7-day window for "last week".
- [OBSERVED] **Language-independent failure mechanisms dominate absolute failure counts:** benchmark defects (31.5% of gemma English failures on OfficeBench), malformed actions (gpt-oss: 1,671 invalid actions in English on OfficeBench), wrong tool arguments (40 of 79 gpt-oss failures on WorkBench), unverified empty results, cut-off output loops, step-budget exhaustion on three-app tasks (gemma 25%, gpt-oss 7.1% in English).
- [INTERPRETATION] **Errors that originate early are rarely recovered.** Across the traced chains, a grounding or planning error is followed by a control failure: the empty result is accepted, the same malformed action is retried, the correct evidence is rejected. The decisive weakness is usually *execution-level control* (verification and recovery), not the initial reasoning step alone.
- [HYPOTHESIS] **gpt-oss's Bangla advantage on OfficeBench is a format effect, not better reasoning.** The Bangla prompt coincides with fewer invalid actions (1,671 → 955) and fewer built-in tool calls (643 → 99); removing English format failures symmetrically shrinks the advantage to −3.3 pp (n.s.). Causality needs a format ablation.
- [OBSERVED] **The two models fail at different layers of the same agent pipeline.** gpt-oss:20b fails early, at the action-format and tool-argument layers (50% of its OfficeBench English failures produce no valid action; 51% of its WorkBench failures are wrong arguments). gemma4:31b gets past those layers and fails later, on content, step budget and the language of its answer (8% no-valid-action failures). [INTERPRETATION] Why: each model's training defaults collide with a different part of the benchmark interface, and a model that fails early never reaches the later layers (Section 4.5).

Highest-value actions. (1) Make checkers language-agnostic (normalise Bengali digits, accept Bangla answers, or state the answer language). (2) Fix the user simulator so it cannot confirm and stop in one turn and must call device tools. (3) Keep names, literals and digits protected in translation (now done for OfficeBench v2: 298/298 tasks pass the automatic checks). (4) Add verification and stuck-recovery to the agent loop (retry lookups with case variants, chunk long writes, change strategy after repeated malformed actions). (5) Run ≥ 3 seeds per cell, because TAC shows run-to-run variation as large as the language gap.

0. Data, design and method

All four studies use a **paired design**: the same task runs once in English and once in Bangla (system prompt and task translated; tool names, IDs, file names and URLs kept English), at temperature 0, one run per cell. Each paired task falls into PP (both pass), PF (English pass, Bangla fail), FP (English fail, Bangla pass) or FF (both fail). The gap is English minus Bangla success; MELR = (EN − BN) / EN. Tests are exact McNemar on discordant pairs, Wilson intervals for rates and paired bootstrap (4,000 resamples) for gaps; Holm correction is applied within families of subgroup tests.

Every hand-analysed failure receives a **first failure point** (the earliest step that, corrected alone, would most likely have passed), a **failure type** and one **confound class** from a shared seven-class scheme, decided by the same decision tree (translation → simulator → understanding → tool → argument → environment → verification), with an added Step 0: “is the output already correct for the task as written?”.

Class	Meaning	Side of the system
C1 Genuine Bangla agent failure	The Bangla input led the agent astray; English did not fail this way	Agent
C2 User simulator / NPC error	The Bangla simulated user/coworker stopped early, gave wrong values, replied in English, skipped device actions	Benchmark (simulator)
C3 Translation / localization	The translated text changed the task or a literal; or localized NPC text breaks a check	Benchmark (task)
C4 Tool / environment	A harness, tool or infrastructure fault blocks a valid action	Harness
C5 Checker / task ambiguity	The check rejects a correct outcome (e.g., answer in Bangla script) or the task admits two readings	Benchmark (check)
C6 General model weakness	The same mechanism fails in an English run of the same model	Agent (language-independent)
C7 Unclear / noise	No visible mechanism, or reruns of the same task disagree	—

Table 0.1. Confound classes shared by all four analyses. C2 is structurally zero on OfficeBench and WorkBench (no simulated user).

Outcome categories used in this report

- **Complete task failure**: no required artefact or write exists at the end (e.g., stuck runs, no file, write never called).
- **Partial success**: some checkpoints or sub-goals met (TAC partial points; compound tasks with one part missing).
- **Inefficient execution**: the run wastes its budget on invalid or repeated actions, whether or not it eventually passes.
- **Incorrect or incomplete output**: an artefact or answer exists but its content, format or script is wrong.

Reliability of the labels

Benchmark	Annotation	Agreement
OfficeBench	Single annotator for the 42 + 25 trace notes; codebook validated on 16 double-coded cases	$\kappa = 0.78$ primary code, 0.64 side (16 cases)
τ^2-bench	46 of 66 notes double-coded and adjudicated; 20 newest single-annotator	$\kappa = 0.39$ fine confound; 0.71 artefact vs agent
WorkBench	Annotator, blind second reviewer and adjudicator on every EN-pass/BN-fail case (all Claude models)	Confound κ: gemma 0.59, gpt-oss 0.77
TAC	Single annotator (Claude model), 8 cases	Not measured

Table 0.2. Label provenance. All annotators are LLMs; native-speaker review of translations is pending in WorkBench.

1. Results and analysis

1.1 Overall success, English vs Bangla

Benchmark	Model	n	EN	BN	Gap (pp) [95% CI]	MELR	McNemar p
OfficeBench	gemma4:31b	300	50.3%	42.3%	+8.0 [3.0, 13.0]	15.9%	0.003
OfficeBench	gpt-oss:20b	300	28.7%	35.3%	−6.7 [−12.0, −1.3]	−23.3%	0.022
τ²-bench	gemma4:31b	218	49.5%	38.1%	+11.5 [5.0, 17.4]	23.1%	0.001
τ²-bench (teammate set)	gemma4:31b	219	33.8%	25.1%	+8.7	25.7%	0.034
τ²-bench	gpt-oss:20b	149	38.9%	39.6%	−0.7 [−10.7, 8.7]	−1.7%	1.000
WorkBench	gemma4:31b	90	81.1%	82.2%	−1.1 [−10.0, 7.8]	−1.4%	1.000
WorkBench	gpt-oss:20b	90	58.9%	53.3%	+5.6 [−5.6, 16.7]	9.4%	0.424
TAC (latest BN run)	gemma4:31b	23	39.1%	26.1%	+13.0 [0.0, 26.1]	33.3%	0.250

Table 1.1. Success rates by benchmark, model and language. Positive gap = Bangla worse. τ² denominators exclude infrastructure-error pairs on both sides; gpt-oss retail and banking lost many tasks to provider errors.

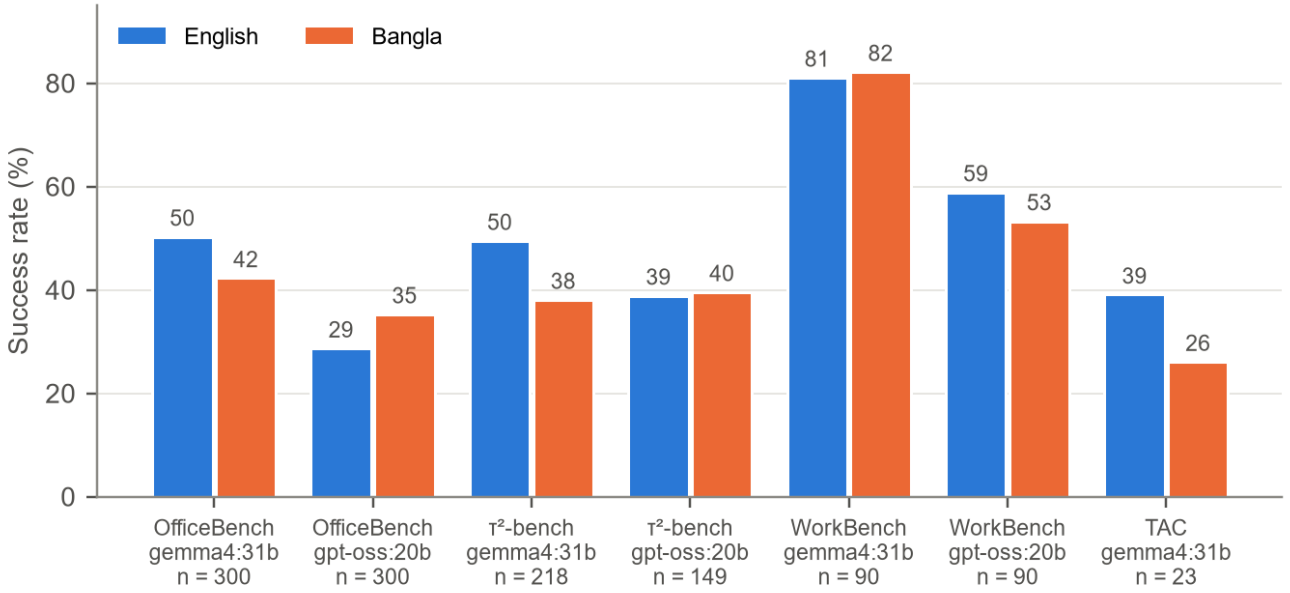


Figure 1.1. Success rate of every run. Model differences are larger than language differences: gemma leads gpt-oss by 21.6 pp (OfficeBench), 10.6 pp (τ², different denominators) and 22.2 pp (WorkBench) in English.

[OBSERVED] Three of the seven model-benchmark comparisons show a significant gap, in **opposite directions**: gemma loses in Bangla on OfficeBench and τ²-bench; gpt-oss gains in Bangla on OfficeBench. The difference between the two models' gaps is +14.7 pp [7.7, 22.3] on OfficeBench and +14.1 pp [2.0, 26.2] on the 149 τ² tasks shared by all four cells, but −6.7 pp [−20.0, 6.7] on WorkBench.

[INTERPRETATION] A single claim of the form "agents are worse in Bangla" is not supported: the sign of the language effect depends on the model, and its size depends on which benchmark components (checker, simulator, translation) see the Bangla text.

1.2 Paired outcomes

Benchmark · model	PP	PF (EN only)	FP (BN only)	FF	Discordant balance
OfficeBench · gemma	109	42	18	131	2.3 : 1 against Bangla
OfficeBench · gpt-oss	61	25	45	169	1 : 1.8 in favour of Bangla
τ²-bench · gemma	68	40	15	95	2.7 : 1 against Bangla
τ²-bench · gpt-oss	32	26	27	64	balanced
WorkBench · gemma	66	7	8	9	balanced
WorkBench · gpt-oss	38	15	10	27	1.5 : 1 against Bangla (n.s.)
TAC · gemma	6	3	0	14	one-directional, 3 pairs

Table 1.2. Paired outcome counts. FF dominates OfficeBench and TAC: most tasks that fail, fail in both languages.

[OBSERVED] Flips occur in both directions everywhere except TAC. Even where gemma is significantly worse in Bangla, 18 (OfficeBench) and 15 (τ²) tasks pass *only* in Bangla. [INTERPRETATION] At one run per cell, part of every discordant count is sampling noise; TAC's reruns (Section 1.5) show how large that part can be.

1.3 Task size and domain

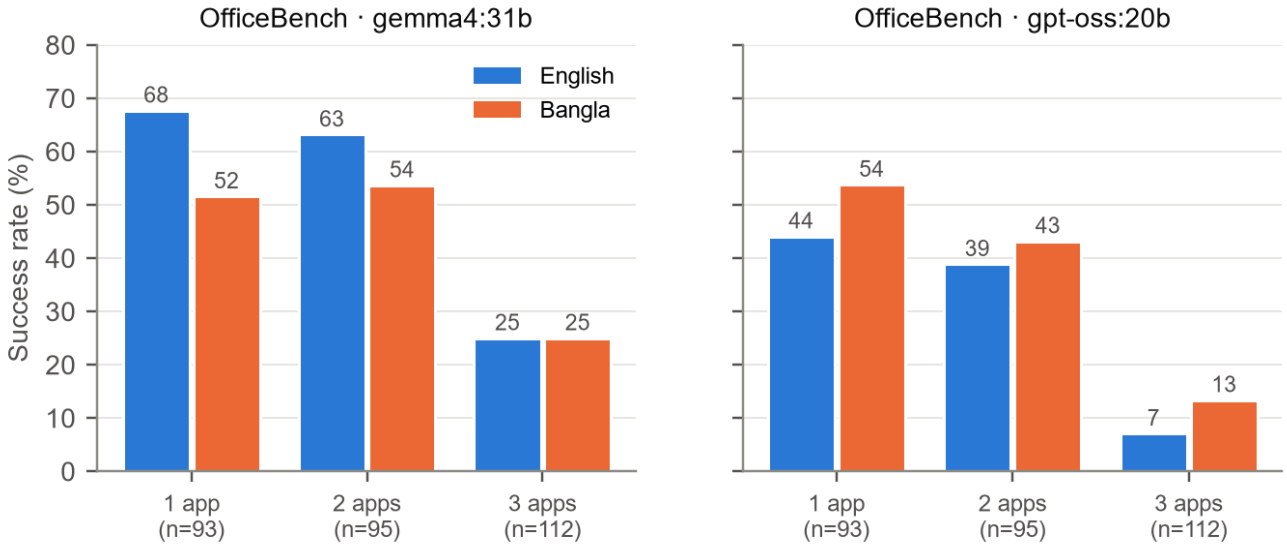


Figure 1.2. OfficeBench success by number of apps. gemma's Bangla gap is concentrated in short tasks; gpt-oss is higher in Bangla at every size; every run collapses on three-app tasks.

[OBSERVED] On OfficeBench the gemma gap is +16.1 pp on single-app tasks (Holm-adjusted p = 0.004), +9.5 pp on two-app (adjusted p = 0.186) and exactly 0 on three-app tasks (28/112 in both languages). Every run falls sharply with task size: gemma English 67.7% → 25.0%, gpt-oss English 44.1% → 7.1%. [INTERPRETATION] Three-app tasks fail for the same reasons in both languages (step budget, cut-off writes, benchmark defects), which masks any language effect; the zero gap there says nothing about Bangla on hard tasks.

τ²-bench domain	gemma EN	gemma BN	gemma gap	gpt-oss EN	gpt-oss BN	gpt-oss gap
Airline	66.0%	46.0%	+20.0	39.6%	45.8%	−6.2
Retail	73.2%	51.2%	+22.0	43.9%	26.8%	+17.1
Telecom	47.1%	41.4%	+5.7	39.1%	50.0%	−10.9
Banking	10.0%	7.5%	+2.5	21.4%	21.4%	0.0

Table 1.3. τ²-bench by domain. No single domain survives Holm correction. Banking is near floor for both models and carries no language signal.

WorkBench domain	gemma EN	gemma BN	gpt-oss EN	gpt-oss BN
Email	100.0%	93.3%	73.3%	66.7%
Calendar	86.7%	93.3%	66.7%	73.3%
Analytics	86.7%	66.7%	73.3%	46.7%
Project mgmt	86.7%	93.3%	53.3%	60.0%
CRM	73.3%	86.7%	60.0%	46.7%
Multi-domain	53.3%	60.0%	26.7%	26.7%

Table 1.4. WorkBench by domain (15 tasks each; descriptive only). Analytics shows the largest one-way drop for both models; multi-domain is hardest in both languages.

TAC run	Full completion	Set 1 (RocketChat, GitLab)	Set 2 (ownCloud)	Mean partial score
English	9/23 (39.1%)	6/11 (54.5%)	3/12 (25.0%)	0.515
Bangla, latest run	6/23 (26.1%)	5/11 (45.5%)	1/12 (8.3%)	0.433
Bangla, first run	5/23 (21.7%)	4/11 (36.4%)	1/12 (8.3%)	0.423

Table 1.5. TAC full completion and partial credit. ownCloud tasks (Set 2) are hardest in both languages.

1.4 Unexpected outcomes

- [OBSERVED] **gpt-oss improves in Bangla on OfficeBench.** 45 subtasks pass only in Bangla vs 25 only in English; 19 of the 45 English failures are action-format failures, and all 45 English runs contain at least one malformed action.
- [OBSERVED] **gemma's Bangla runs are leaner on WorkBench** (3.29 requests and 3.17 tool calls per task vs 3.78 and 3.96 in English), largely because two English analytics runs looped on the same read call.
- [OBSERVED] **Some English passes are not real successes.** WorkBench analytics:98 passed in English by abstaining after 24 looping read calls; OfficeBench 1-2/4 passes in English only because its answer happens to contain the keyword "Tom".
- [OBSERVED] **Correct answers fail.** On OfficeBench 1-2/0 all four cells give the right free slot (14:30–15:30) and all four fail, because two of the four checks require a calendar event the task never asks for.

1.5 How much of the gap survives confound filtering

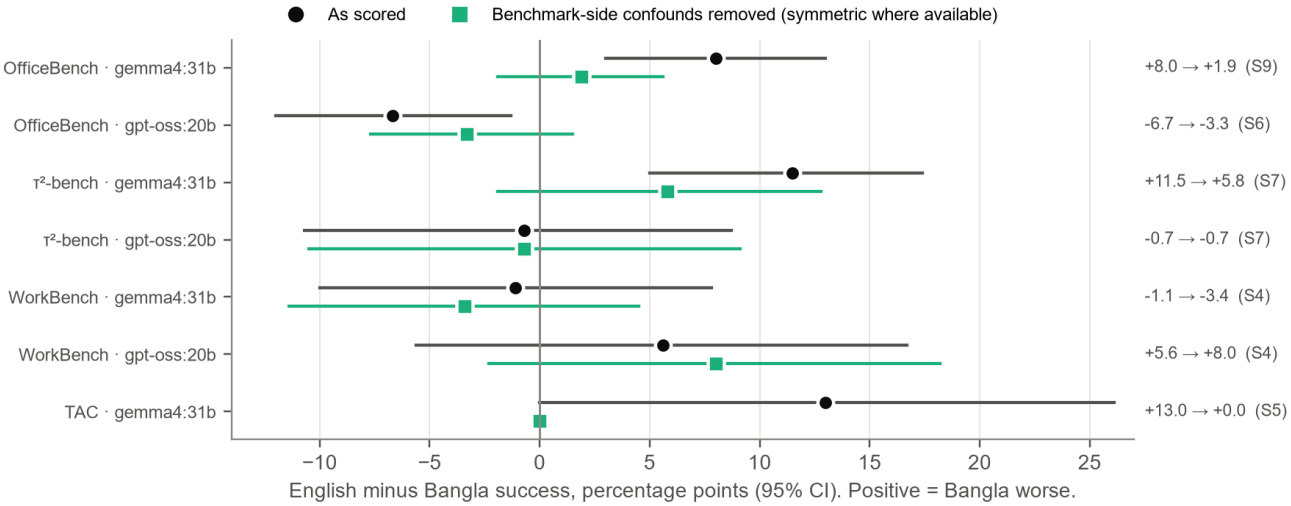


Figure 1.3. English − Bangla gap as scored (black) and after removing benchmark-side confounds (green). Scenario codes refer to the source reports (OfficeBench S9: drop translation, tool and checker cases on both sides; τ² S7: drop premature-stop dialogues in either language and credit Bengali-digit answers; WorkBench S4: drop translation, tool and checker cases symmetrically; TAC S5: credit the checker artefact and drop tasks whose Bangla reruns disagree).

[OBSERVED] Every significant gap moves toward zero and its interval then includes zero. Crediting correct Bangla-script answers alone moves OfficeBench gemma from +8.0 to +3.0 pp ($p = 0.233$). For τ² gemma, dropping premature-stop dialogues in either language gives +7.1 pp ($p = 0.090$). For TAC, the choice between the first and latest Bangla run moves the gap from +13.0 to +17.4 pp on its own, and 1 of 13 reruns changed full completion (3 changed the score).

[INTERPRETATION] The measured language gaps are of the size that benchmark-side artefacts can explain. This does **not** prove Bangla has no cost: the residual cases (C1 + C7) are too few for one-run-per-cell data to establish or exclude a modest effect. [OBSERVED] One-sided filters are biased: on τ², applying the simulator filter only to EN-pass/BN-fail cases turns gpt-oss's null gap into a −12.8 pp "Bangla advantage", which is why symmetric scenarios are preferred above.

2. Why and where the models fail

2.1 Which runs and tasks fail

Benchmark	Run	Failed runs	Fail in every cell	Pass in every cell
OfficeBench	gemma EN / BN	149 / 173	113 of 300 subtasks fail in all four runs	48
OfficeBench	gpt-oss EN / BN	214 / 194		
τ²-bench	gemma EN / BN	110 / 135	33 of 149 shared tasks fail in all four	21
τ²-bench	gpt-oss EN / BN	91 / 90		
WorkBench	gemma EN / BN	17 / 16	6 of 90 tasks fail in all four	37
WorkBench	gpt-oss EN / BN	37 / 42		
TAC	gemma EN / BN	14 / 17	14 of 23 task-runs fail in both languages	6

Table 2.1. Failure volume. Hard tasks are hard for everyone: the all-cells-fail sets are the largest single group on OfficeBench and TAC.

- [OBSERVED] **Weakest conditions:** three-app OfficeBench tasks (gpt-oss English 7.1%), τ² banking (≤ 21.4% for every cell), TAC ownCloud tasks (Bangla 8.3%), WorkBench multi-domain (gpt-oss 26.7% in both languages).
- [OBSERVED] **Model decides more than language:** 8 WorkBench tasks pass for gemma in both languages and fail for gpt-oss in both; 11 τ² tasks pass in English and fail in Bangla for *both* models (the strongest cross-model candidates for a Bangla-side cause).
- [OBSERVED] **Benchmark defects decide many outcomes:** on OfficeBench a defect decides at least one of three fully coded runs on 57 subtasks; 44 subtasks fail for the identical coded cause in every run.

2.2 Where in the trajectory failures start

The first failure point is grouped into the stage of the trajectory where the decisive error happened: task-level reasoning (instruction understanding, entity/slot grounding, constraint preservation, planning), execution-level action (tool selection, tool-argument construction, tool execution), control (result verification, premature termination) and final response only.

First failure point	OB gemma (42)	OB gpt-oss (25)	τ² gemma (40)	τ² gpt-oss (26)	WB gemma (7)	WB gpt-oss (15)	TAC (8)
Instruction understanding	5	4	0	3	1	0	0
Entity / slot grounding	5	7	7	5	2	1	0
Task constraint preservation	3	1	4	3	1	1	0
Planning	5	1	6	5	1	0	2
Tool selection	0	0	6	1	0	1	0
Tool-argument construction	4	1	4	2	2	11	1
Tool execution	2	2	0	0	0	0	1
Result verification	2	4	1	3	0	0	2
Premature termination	1	2	11	1	0	1	1
Final response only	15	3	1	0	0	0	0
Unclear	0	0	0	3	0	0	1

Table 2.2. First failure point of the analysed EN-pass/BN-fail cases (TAC: the 8 selected cases, which include reruns and both-fail tasks). OB = OfficeBench, WB = WorkBench.

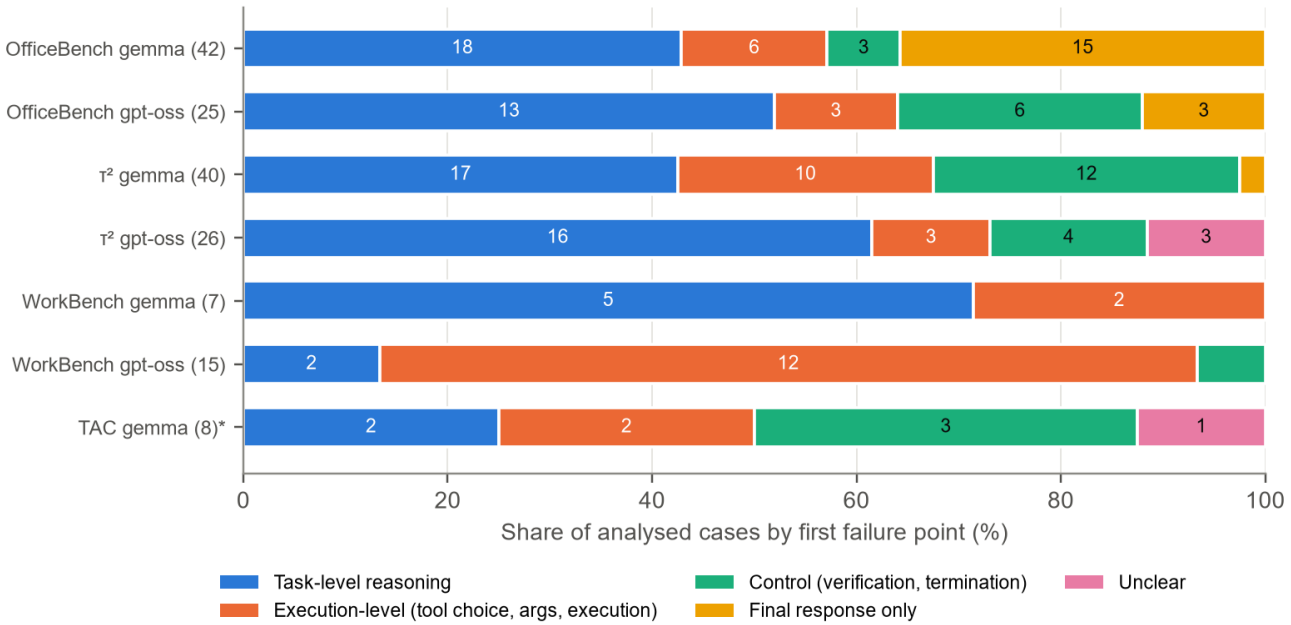


Figure 2.1. Stage at which the decisive error occurs. Note: 10 of τ² gemma's 11 "premature termination" cases were caused by the simulated user, not the agent.

[OBSERVED] Each benchmark has a characteristic failure stage: OfficeBench gemma at the **final response** (15/42: correct content, wrong script); τ² at **task-level grounding/planning** and **termination**; WorkBench gpt-oss at **tool-argument construction** (11/15); TAC at **control** (verification, stopping). [OBSERVED] Tool syntax is never the first failure point of a τ² Bangla failure, and no WorkBench tool call contains Bangla characters (0 of 641 arguments).

2.3 Failure outcome categories

Category	Where it shows up (evidence)	Scale
Complete task failure	OfficeBench stuck runs after empty replies (2-1/0, 2-1/1); cut-off write loops ending with no file (1-15/0, 1-15/1, 2-6/0, 2-22/1); τ² required write never called (gemma 39 EN / 37 BN failures) or the dialogue stopped before the write (26 / 46); TAC admin-remove-pages-pdf produced no file in either Bangla run.	Largest category on τ² and in OfficeBench three-app tasks
Partial success	TAC: 17 of 23 task-runs earn identical partial points in both languages; mean partial score 0.515 EN vs 0.433 BN (Wilcoxon p = 0.141); pm-send-notification 3/4 (one of three people not notified); ml-grade-exam 1/8. OfficeBench compound tasks with a skipped part (3-15/0 image skipped) and dropped columns (1-11/0, 1-6/4).	Only TAC measures it directly
Inefficient execution	gpt-oss OfficeBench: 1,671 invalid actions EN vs 955 BN; 643 vs 99 built-in tool calls; 139 vs 49 steps with no action; 86 vs 63 runs at the 20-step limit. gemma: 14–15 identical rejected writes per TL case. WorkBench analytics:98 EN: 24 looping read calls. gpt-oss WorkBench: 4.23 EN vs 3.84 BN requests per task.	Pervasive for gpt-oss; episodic for gemma
Incorrect or incomplete output	OfficeBench answer in Bangla script/digits (15 gemma cases, e.g., 40 and $20,000 written in Bengali digits); word re-rendering ("Trip", "curd", "08:00 AM"); wrong content (WC: 36 of 131 both-fail gemma Bangla runs); WorkBench wrong filters (crm:77 missing status filter → 5 spurious updates); τ² wrong write arguments (19 per gemma run).	Largest category among Bangla-only failures

Table 2.3. The four outcome categories with representative evidence. The same run can be inefficient and still pass.

2.4 How runs end

OfficeBench gpt-oss:20b (299 runs per language)	English	Bangla
Invalid (malformed) actions	1,671	955
Calls to its own built-in tools (container.exec, …)	643	99
Steps with no action at all	139	49
Runs ended as stuck	31	14
Runs ended at the 20-step limit	86	63
Runs that finished with an answer	182	222

Table 2.4. gpt-oss on OfficeBench: execution-control signals. The Bangla prompt coincides with far fewer format failures.

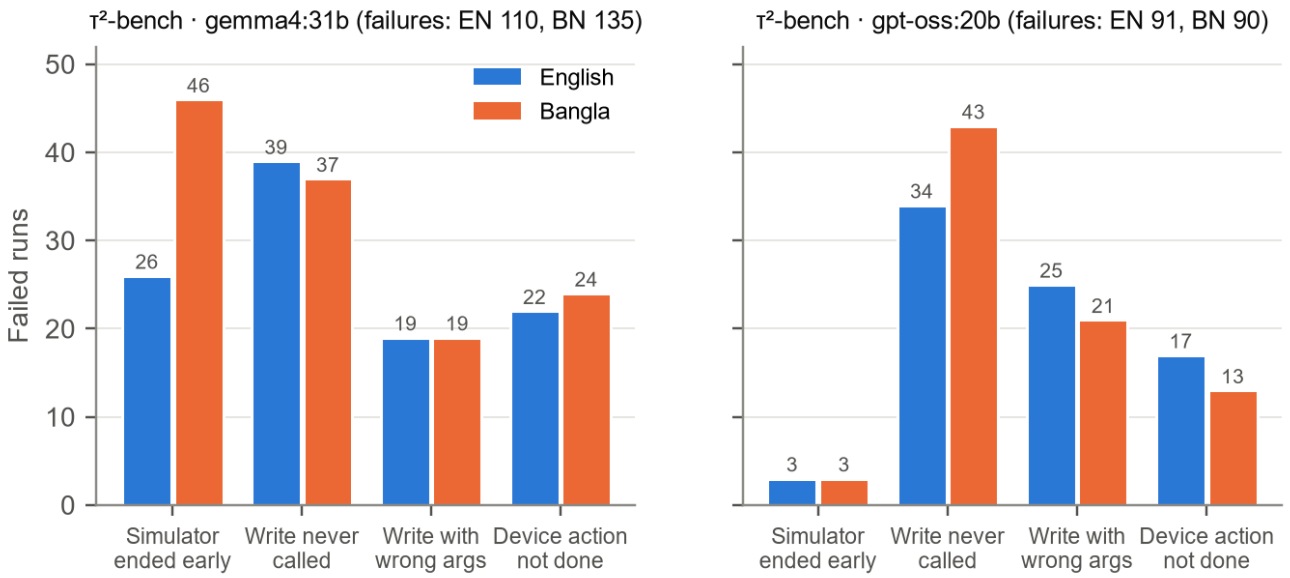


Figure 2.2. τ²-bench: automatic failure signal of every failed run (four largest signals). The only signal that grows sharply in Bangla is the simulator ending the dialogue before the agent acted (gemma 26 → 46).

τ²-bench dialogue signals	gemma EN	gemma BN	gpt-oss EN	gpt-oss BN
Runs where the user writes mostly Latin script	218	33	149	54
Messages per run (mean)	34.70	33.50	40.10	38.80
Agent tool calls per run (mean)	7.69	6.64	7.26	6.44
Runs ending in transfer to a human	108	112	52	55
Simulator ended chat before the agent acted	26	46	3	3

Table 2.5. τ²-bench run-level signals. 33 gemma and 54 gpt-oss Bangla runs have a simulated user writing mostly English, a simulator-fidelity problem the agent cannot fix. The premature-stop detector is heuristic (about half of a random sample of its flags were true premature stops).

3. Root-cause analysis

3.1 Confound classes across benchmarks

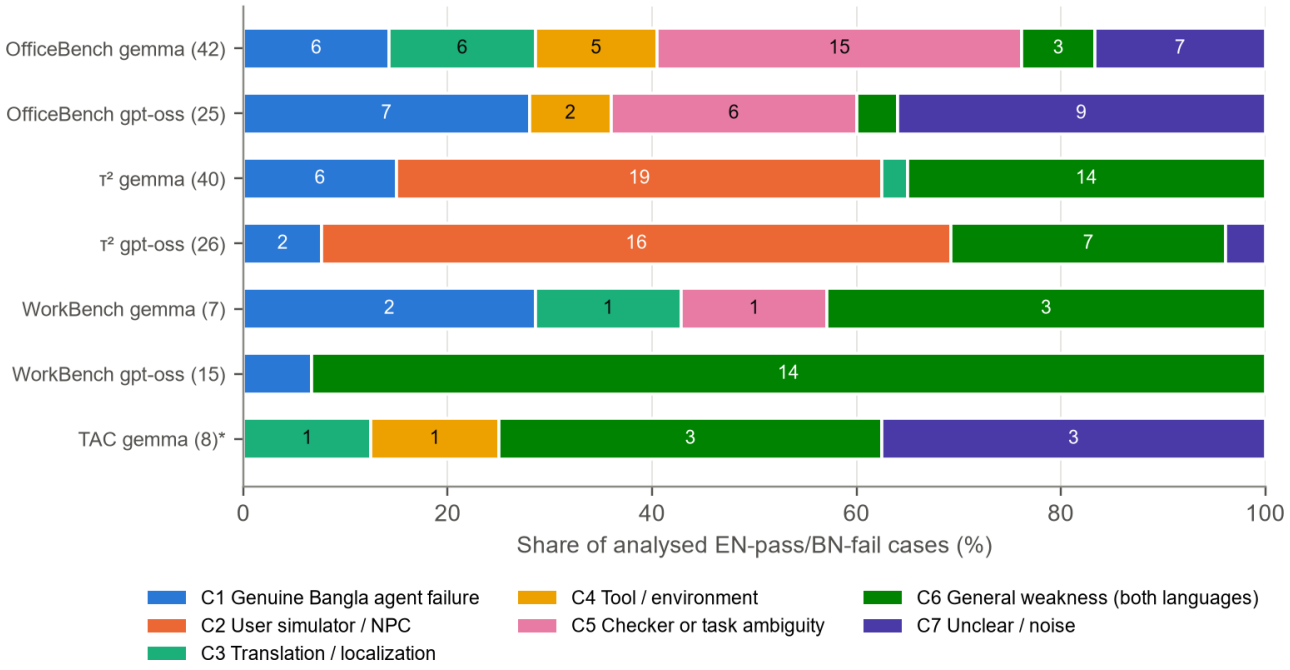


Figure 3.1. Confound class of every hand-analysed case. *TAC's 8 cases include reruns and both-fail tasks; the others are EN-pass/BN-fail cases only.

Class	OB gemma	OB gpt-oss	τ² gemma	τ² gpt-oss	WB gemma	WB gpt-oss	TAC	Total (163)
C1 Genuine Bangla agent failure	6	7	6	2	2	1	0	24 (14.7%)
C2 User simulator / NPC	0	0	19	16	0	0	0	35 (21.5%)
C3 Translation / localization	6	0	1	0	1	0	1	9 (5.5%)
C4 Tool / environment	5	2	0	0	0	0	1	8 (4.9%)
C5 Checker / task ambiguity	15	6	0	0	1	0	0	22 (13.5%)
C6 General weakness	3	1	14	7	3	14	3	45 (27.6%)
C7 Unclear / noise	7	9	0	1	0	0	3	20 (12.3%)
Cases	42	25	40	26	7	15	8	163

Table 3.1. Confound counts. Upper bound on genuine Bangla agent effects (C1 + C7): 44 of 163 (27.0%); lower bound (C1): 24 (14.7%).

[INTERPRETATION] The benchmark's interaction design predicts its dominant confound. Tasks whose answer is a free-text string checked by keyword (OfficeBench) are dominated by checker language; dialogue tasks with a same-model simulator (τ²) by the simulator; strict function-calling tasks with English tool descriptions (WorkBench) by language-independent argument errors; long-horizon tasks with brittle web apps (TAC) by environment walls and run-to-run noise.

3.2 Root causes, mechanisms and contribution to failure

Each cause is listed with its mechanism, how it turns into the final failure, the evidence and its status. **Confirmed** = the mechanism is visible in traces and the counterfactual (usually the English run or a rerun) succeeds; **Plausible** = consistent with traces but with a competing explanation; **Hypothesis** = inferred from aggregate association only.

#	Root cause (category)	Mechanism → contribution to failure	Evidence	Status
R1	English-only checker; answer in Bangla script (benchmark: check)	Prompt never names an answer language → agent answers in Bangla script/digits → keyword check for ASCII value fails although content is correct.	OB: 15 gemma C5 cases (1-10/0 and 2-31/0: numbers in Bengali digits; 1-13/2: Bangla "na" for No); 2 gpt-oss. TAC T01: NPCs reply in Bangla, CP5 needs "helpful". τ²: Bengali-digit fix flips 2 gemma runs.	Confirmed

#	Root cause (category)	Mechanism → contribution to failure	Evidence	Status
R2	Benchmark defects (benchmark: check)	Check asks for something the task does not (or the reference/app is wrong) → correct work scored FAIL in every language.	OB: 47 of 149 gemma EN failures (31.5%), 39 of 131 both-fail; 1-2/0 fails in all four cells; 57 subtasks decided by a defect.	Confirmed
R3	User-simulator errors (benchmark: simulator)	Bangla simulated user confirms and emits ###STOP### in one turn, gives an internal ID as a name, answers in English, or narrates a device action without calling the tool → the required write or device state is never reached.	τ^2: 35 of 66 cases (53%); premature-stop flags 46 BN vs 26 EN (gemma); airline 19; telecom device-action failures.	Confirmed per case; detector counts heuristic
R4	Translation changed the task (benchmark: task)	Machine translation altered a literal or the operation → agent faithfully executes the wrong task.	OB: 3-56/0 quoted homework sentence translated; 3-68/0 “sum” became “add” (wrote 104.98 and 1389.99 instead of 1494.97); 2-38/2 and 3-17/0 files-to-create presented as existing; 3-55/0, 2-40/2. TAC: translations faithful in all 8 cases.	Confirmed
R5	Entity grounding from caseless script (reasoning: grounding)	Name present only in Bangla script → agent writes it in lower case → case-sensitive mailbox/calendar lookup returns empty → empty result accepted (see R12).	OB gemma 1-20/3, 1-21/1, 1-21/2; gpt-oss 1-21/1, 1-21/2, 2-39/0, 2-39/1, 2-39/3, 2-40/1 (same two subtasks fail for both models). τ^2: Bangla-script names/IDs in lookups.	Confirmed; replicated across models
R6	Re-rendering values into English (reasoning: argument construction)	Agent maps a Bangla word or time back to a different English surface form → exact-match check fails.	OB 1-1/4 “Trip to Florida” (both models), 1-4/1 “curd” for yogurt, 1-6/2 “08:00 AM” for 8:00.	Confirmed per case; medium confidence it is l anguage-caused
R7	Relative-date grounding (reasoning: grounding)	“Last week” grounded as rolling 7 days instead of the previous calendar week → searches return nothing → correct evidence later rejected.	WB email:41 (gemma): windows 11-23..29 vs 11-19..26; undated search found both emails, agent rejected them.	Plausible (pending native-speaker review of the phrase)
R8	Malformed actions / format non-adherence (execution: tool-call format)	Model emits its native tool-call habit (container.exec, repo_browser), invalid JSON or empty replies → harness rejects the step → repeated invalid steps end the run as stuck or at the step limit.	OB gpt-oss: 54 FMT failures; 1,671 invalid actions EN; 19 of 45 FP subtasks are EN format failures. Pilot qwen2.5-7b: 7 of 12 runs stuck. WB: invented argument names (new?, field, value_to_plot) crash runs (5 BN, 2 EN).	Confirmed as mechanism; the link to prompt language is a Hypothesis
R9	Output cap + whole-document rewrite (execution control; context/output limits)	Agent plans to resend the entire file (write tool’s append behaviour undocumented) → action exceeds the 512-token reply cap → “Malformed action!” → identical retries until the step budget is spent.	OB 1-15/0 (14 rejected writes), 1-15/1 (15), 2-6/0, 2-22/1 (English recovered by chunking in 2-6/0).	Confirmed
R10	Wrong tool arguments (execution: argument construction)	Missing or wrong filter values, thresholds or IDs → wrong rows changed or nothing found.	WB gpt-oss: 40 of 79 labelled failures; crm:77 status filter dropped → 5 spurious updates; τ^2 “write with wrong arguments” 19/19 (gemma), 25/21 (gpt-oss); τ^2 airline 8 invented 2 free bags.	Confirmed; lang uage-independe nt (C6)
R11	Tool-selection mistakes (execution: tool choice)	Agent calls tools that belong to the user or do not exist, or transfers instead of using a discoverable tool → required action never happens.	τ^2 telecom: run_speed_test / toggle_airplane_mode called by the agent → “not found”; banking: transfer instead of unlock_discoverable_agent_tool; WB crm:77 non-existent customer_relationship_manager _update_task.	Confirmed
R12	No verification of results (control)	Empty or implausible tool results accepted as final; success claimed without checking state.	OB 1-20/2 (empty listing accepted at step 2; English searched the filesystem), ER code 9 both-fail runs; TAC hr-transfer-group false success claims, sde-install-go verified by absolute path (grader’s `go version` fails), ml-grade-exam guessed grades.	Confirmed
R13	Premature termination (control)	Agent stops at the first acceptable answer or before finishing all sub-goals.	TAC hr-pick-interviewer-2 (both languages stop after first acceptance), pm-send run A (2 of 3 notified); OB gpt-oss 1-2/3 “I don’t know” after one step; τ^2 early transfer.	Confirmed

#	Root cause (category)	Mechanism → contribution to failure	Evidence	Status
R14	Step budget exhaustion on long tasks (control; task length)	Three-app tasks need more steps than 20 once one retry loop occurs → budget runs out.	OB SB code 18 of 131 both-fail gemma; three-app success 25.0% (gemma) / 7.1% (gpt-oss EN).	Confirmed for three-app tasks
R15	Harness / environment faults (harness)	llm app limited to 50 tokens returns empty strings for a reasoning model; empty model replies; shell quoting; ownCloud 401/permission walls; admin password prompts; provider errors.	OB gpt-oss 2-36/0, 3-89/0 (all 14 llm calls empty); 2-1/0, 2-1/1; 3-9/0 apostrophe; TAC T05 (401), T07 (password modal); WB provider errors (S6 leaves gap unchanged).	Confirmed mechanism; Bangla-triggered empty replies a Hypothesis
R16	Context handling (harness: memory)	Earlier thoughts are not carried forward in the OfficeBench loop (only actions/results are) → the agent may re-derive or forget a plan and repeat a failed action.	Design note in the OfficeBench report; repeated identical retries in R8/R9 are consistent with it.	Hypothesis
R17	Run-to-run variance (measurement)	Single-run attributions mix mechanism and sampling noise.	TAC: 1 of 13 reruns changed full completion, 3 changed score; sde-install-go passes and fails in Bangla. OB C7: 2-32/1, 3-87/0 diverge on identical OCR input.	Confirmed as variance; cause per case unknown

Table 3.2. Root-cause catalogue. OB = OfficeBench, WB = WorkBench. Categories map to the requested taxonomy: reasoning errors (R5–R7), tool-selection mistakes (R11), malformed tool calls (R8), failure to verify (R12), repeated unsuccessful retries and inefficient execution (R8, R9), context/output limitations (R9, R16), premature termination (R13), plus benchmark-side (R1–R4) and harness (R15) causes.

3.3 How much each cause explains

Cause	Effect on the measured result	Evidence type
R1 answer-language checker	OfficeBench gemma gap +8.0 → +3.0 pp when Bangla-script answers are credited	[OBSERVED]
R2–R4 checker, translation, harness	OfficeBench gemma gap → −0.7 pp (one-sided) / +1.9 pp (symmetric S9)	[OBSERVED]
R3 user simulator	τ^2 gemma gap +11.5 → +7.1 pp after dropping premature-stop dialogues in either language	[OBSERVED]
R8 format failures	OfficeBench gpt-oss Bangla advantage −6.7 → −3.3 pp after dropping English format failures	[OBSERVED] [HYPOTHESIS]
R1 + R17 noise (TAC)	TAC gap +13.0 → 0.0 pp after crediting the checker artefact and dropping disagreeing reruns	[OBSERVED]
R10–R14 general weaknesses	Dominate absolute failure counts in every benchmark (e.g., 82 of 112 WorkBench failed runs are C6)	[OBSERVED]

Table 3.3. Contribution of each cause family to the headline numbers.

3.4 Confirmed causes vs open hypotheses

- **Confirmed** (trace-level mechanism plus a passing counterfactual): R1, R2, R3 (per case), R4, R5, R9, R10, R11, R12, R13, R14, R15 (mechanism).
- **Plausible** (mechanism visible, alternative not excluded): R6 word re-rendering (could be lexical ambiguity of the Bangla word); R7 relative-date grounding (could be a translation that leans to “past 7 days”, which would make it C3).
- **Hypotheses** (aggregate association only): Bangla prompts suppress gpt-oss’s native tool-call habit (R8); empty replies in 2-1/0 and 2-1/1 are Bangla-triggered (R15); missing thought carry-over causes repeated retries (R16); the 9 gpt-oss and 7 gemma C7 OfficeBench cases hide a genuine Bangla reasoning cost.
- **Not supported by the data**: “Bangla breaks tool calling” (no case fails first on tool syntax; 0 Bangla characters in 641 WorkBench arguments) and “Bangla agents reason worse” in general (no EN-pass/BN-fail OfficeBench case has a wrong number, date or path tied by the trace to reading Bangla).

4. Comparative and trajectory-level analysis

4.1 Across models

Dimension	gemma4:31b	gpt-oss:20b
English capability	Higher on every benchmark (50.3 / 49.5 / 81.1 / 39.1%)	Lower (28.7 / 38.9 / 58.9%)
Bangla effect	Cost on OfficeBench (+8.0) and τ² (+11.5); none on WorkBench	Gain on OfficeBench (−6.7); none on τ² and WorkBench
Dominant own failure	Content and planning errors; cut-off write loops; answers in Bangla script	Action-format failures and wrong tool arguments
Answer language	Answers in Bangla when prompted in Bangla (15 C5 cases on OB)	Mostly answers in English (4 Bangla answers among 25 cases)
Shared Bangla mechanisms	Name case (3 OB), word re-rendering (Trip, curd, 08:00 AM)	Name case (6 OB), word re-rendering (Trip)
Tool-related harness issue	512-token reply cap	llm app 50-token limit consumed by hidden reasoning

Table 4.1. Model comparison. The same Bangla mechanism appears in both models; their different net effects come from model-specific failure modes that the Bangla prompt amplifies (gemma: script of the answer) or dampens (gpt-oss: format).

4.2 Across benchmarks and languages

Benchmark	Where Bangla text reaches a brittle component	Resulting dominant confound	Net gap (gemma)
OfficeBench	Free-text answers checked by English keywords; machine-translated literals	C5 checker, C3 translation	+8.0 pp, significant
τ²-bench	The simulated customer itself speaks Bangla and controls when the dialogue ends	C2 simulator	+11.5 pp, significant
WorkBench	Nowhere critical: English tool descriptions, DB-state check, no free-text answer	C6 general weakness	−1.1 pp, n.s.
TAC	Bangla NPC replies stored verbatim, English keyword checkpoint; small sample	C7 noise, C6, one C3	+13.0 pp, n.s.

Table 4.2. The size of the Bangla gap tracks how many benchmark components consume Bangla text, not the agent's tool use.

[INTERPRETATION] WorkBench is the closest to a controlled test of the agent alone (no simulator, no free-text check), and it shows no gap for either model. This supports the reading that most of the gaps elsewhere are produced by the evaluation pipeline's handling of Bangla rather than by the agents' task competence; the boundary is WorkBench's small size (90 tasks; a cost up to 8–17 points cannot be excluded).

4.3 How early mistakes propagate

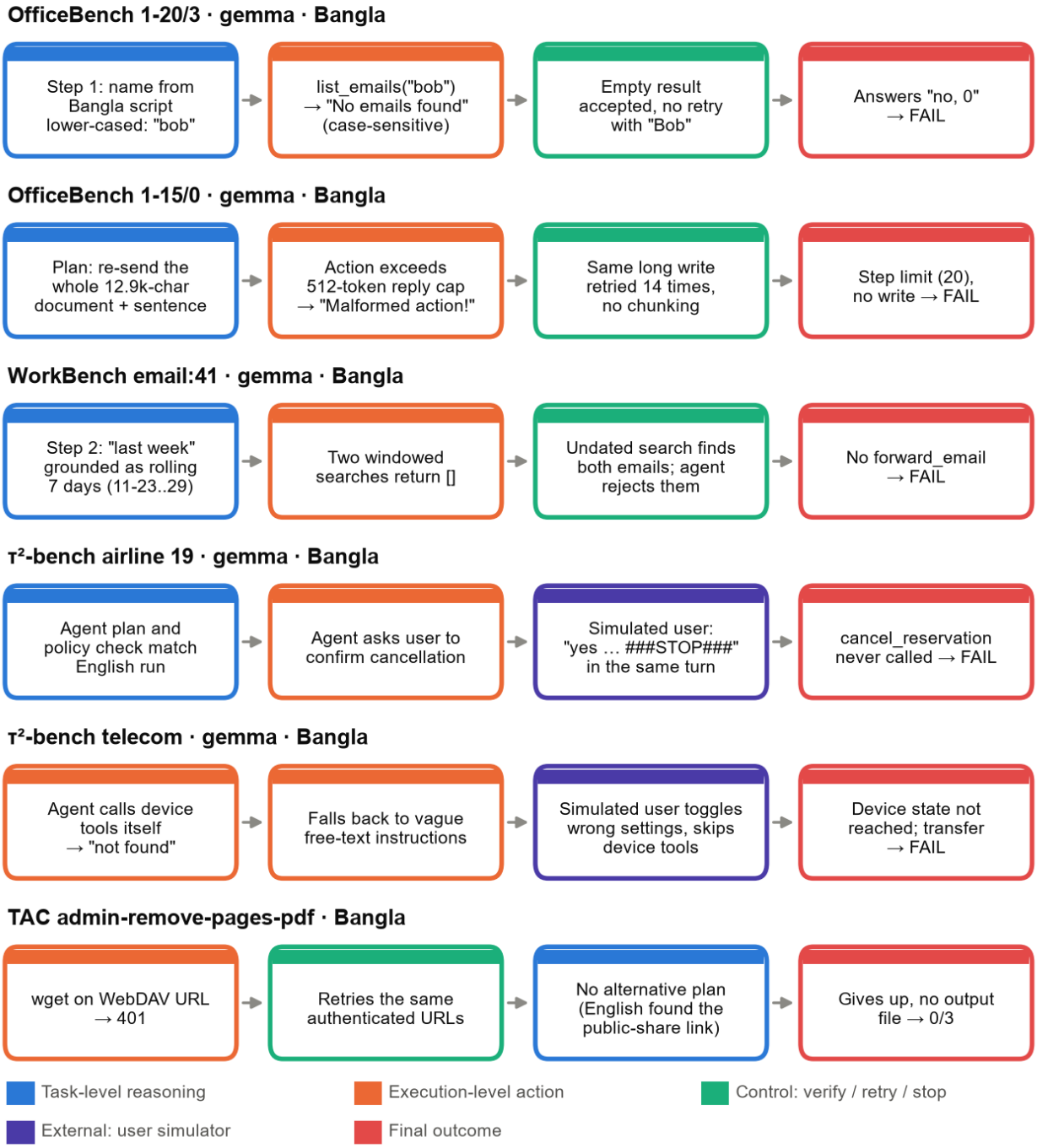


Figure 4.1. Six traced failure chains. Box colour marks the kind of step; each chain is taken from the trace notes in the source reports.

[OBSERVED] Patterns visible in the chains:

- **Grounding → silent empty result → acceptance.** In 1-20/3, a one-character case error makes the lookup return nothing; the agent treats “no emails” as the answer. The English run of 1-20/2 shows the recovery that was missing: after an equally wrong first lookup it searched the filesystem and found the mailbox.
- **Plan → hard limit → identical retries.** In 1-15/0 the plan is reasonable given an under-documented tool, but the agent never adapts after the first rejection; 14 identical actions consume the budget. In 2-6/0 English hit the same wall five times and then recovered by chunking.
- **Early grounding overrides later evidence.** In WorkBench email:41 the wrong date window is chosen at step 2; at step 5 the correct emails are retrieved and *rejected* because they conflict with the agent’s own earlier interpretation.

- **External termination.** In τ² airline 19 the agent's chain is correct to the last step; the simulated user ends the dialogue, so no agent-side change could have passed.
- **Tool misuse → degraded instructions → simulator drift.** In τ² telecom, the agent first calls user-side tools itself, then gives vague instructions; the simulated user toggles the wrong settings, and the run ends in transfer.
- **Environment wall → no alternative plan.** In TAC admin-remove-pages-pdf both Bangla runs retry the same authenticated URLs after a 401; English found a public-share-link workaround (and still failed on content).

4.4 Task-level reasoning, execution-level control, or their interaction?

Origin	Typical cases	Share of analysed cases
Task-level reasoning only	Translation-induced misreadings (OB 3-68/0), filtered roster (2-20/0), τ² airline 9 (flown reservation cancelled), dropped constraints	Largest first-failure stage on τ² and OfficeBench gpt-oss (Figure 2.1)
Execution-level control only	Format failures (gpt-oss), wrong tool arguments (WB gpt-oss 11/15), harness faults, cut-off loops	Dominant on WorkBench gpt-oss and OfficeBench gpt-oss English
Interaction	Grounding error not caught by verification (1-20/3, 1-21/x, email:41); reasonable plan defeated by a hard limit without adaptation (1-15/x); tool misuse amplified by simulator (telecom)	Most of the genuine Bangla cases (C1)
Outside the agent	Checker language (C5), simulator (C2), benchmark defects	35.7% of gemma OB cases; 53% of τ² cases

Table 4.3. Origin of failures.

[INTERPRETATION] The genuine Bangla failures share a two-stage structure: a small task-level slip specific to the Bangla input (case, word, date window) that would be harmless if execution-level control verified results. [HYPOTHESIS] Most C1 cases would therefore be removed either by fixing the input (protect names, dates and literals) or by adding verification (retry empty lookups with case variants; re-check a window that returns nothing). Experiment E6 tests this.

4.5 Why the two models fail at different layers

An agent run passes through the same layers in every benchmark: (L1) read and ground the task, (L2) plan, (L3) emit an action in the required format, (L4) fill in the tool arguments, (L5) control the run (check results, recover, decide when to stop), (L6) write the final answer. The question is at which layer each model breaks first.

Layer	gemma4:31b	gpt-oss:20b	Source
L3 Action format (no valid action, malformed or invented tool calls)	12 of 149 English failures (8%)	107 of 214 English failures (50%); 1,671 invalid actions; 643 calls to its own built-in tools	OfficeBench, Fig. 7 and II.9 of source
L4 Tool arguments	11 of 33 failed runs (33%)	40 of 79 failed runs (51%); invented argument names crash runs	WorkBench R.5
L1–L2 Content and task reading (wrong content, task read differently)	43 of 149 English failures (29%) are wrong content	49 of 214 (23%)	OfficeBench, Fig. 7
L5 Control (step budget, cut-off loops)	42 of 149 English failures (28%) run out of steps or loop at the reply cap	24 of 214 (11%)	OfficeBench, Fig. 7
L6 Answer language (correct answer in Bangla script)	15 of 42 EN-pass/BN-fail cases	3 of 25 (only 4 Bangla-script answers among them)	OfficeBench §6, II.8
Shared Bangla input slip (L1: name case)	3 cases	6 cases (same two subtasks fail for both)	OfficeBench F3, G4

Table 4.4. First broken layer by model. OfficeBench shares use the primary root-cause families of all coded failures; WorkBench shares use the adjudicated failure type of every failed run.

Why gpt-oss breaks at the format and argument layers

- [INTERPRETATION] **Its own tool-calling habit competes with the benchmark's format.** OfficeBench asks for plain text ("##Command: {json}"). gpt-oss repeatedly calls tools that exist in its own training setup but not in OfficeBench (container.exec, repo_browser: 643 calls in English). Each such call is rejected as malformed, and enough of them end the run as stuck. [OBSERVED] Evidence: 50% of its English failures produce no valid action; all 45 English runs that Bangla later passed contain at least one malformed action.
- [INTERPRETATION] **When the interface handles the format, the errors move one layer down.** WorkBench uses provider-side function calling, so the JSON syntax is no longer gpt-oss's problem; its failures then concentrate at argument filling (51%): missing filters, wrong thresholds, invented argument names. The model's weakness is turning task constraints into exact structured values, and the interface decides whether that shows up as a format

error or an argument error.

- [INTERPRETATION] **Hidden reasoning consumes fixed token budgets.** gpt-oss reasons in a separate channel before answering. Where the harness allows only 50 tokens (the OfficeBench llm app), the reasoning uses them up and the visible answer is empty (all 14 calls in 2-36/0 and 3-89/0). [OBSERVED] This failure does not occur for gemma.
- [HYPOTHESIS] **Capacity.** gpt-oss:20b is a mixture-of-experts model with far fewer active parameters per token than gemma4:31b; precise argument construction may suffer from that. This is not tested by any of the four studies.

Why gemma breaks at the content, control and answer layers

- [INTERPRETATION] **It follows the text action format reliably** (8% no-valid-action failures), so most of its runs reach the later layers, and its failures appear there: wrong content (29%), running out of steps or looping at the 512-token reply cap (28%), and benchmark defects that only matter once a run produces output (32%).
- [INTERPRETATION] **It mirrors the user's language in its answer.** Given a Bangla prompt, gemma writes its final answer in Bangla script and digits; the English keyword checker then rejects it (15 of its 42 Bangla-only failures). gpt-oss stays anchored to English in its answers, so this layer barely affects it. The same behaviour that makes gemma a better Bangla conversational partner is penalised by an English-only checker.
- [INTERPRETATION] **Its control layer is the weak point under hard limits.** When an action is rejected for length, gemma tends to resend the same long action (14–15 identical retries in 1-15/0 and 1-15/1) instead of changing strategy; this is a recovery failure, not a reasoning failure.

Two cross-cutting reasons

- [INTERPRETATION] **Failure masking.** A run can only fail at a later layer if it survives the earlier ones. gpt-oss loses half its English runs at the format layer, so its content, budget and answer-language failures are under-counted, not absent. Comparing models by failure counts per layer therefore needs per-layer pass rates (how many runs reached each layer), which none of the source reports records yet.
- [INTERPRETATION] **Language interacts with the layer, not the model.** The Bangla input changes behaviour at the layer where each model is fragile: it reduces gpt-oss's format failures (1,671 → 955 invalid actions) and increases gemma's answer-language failures. The one Bangla mechanism located at L1, the lower-cased name, appears in both models, because it comes from the input script (no letter case), not from either model.

[HYPOTHESIS] Prediction that tests this account: give gpt-oss native function calling on OfficeBench and its Bangla advantage should disappear while its argument errors rise (experiment E5); tell gemma to answer in English and its Bangla gap should fall to about +3 pp (E2).

5. Evidence and recommendations

5.1 Evidence matrix for the major findings

Finding	Observed facts	Interpretation	Open hypothesis
Bangla gap is model- and benchmark-dependent	gemma +8.0 (OB, p=.003), +11.5 (τ^2, p=.001), −1.1 (WB); gpt-oss −6.7 (OB, p=.022), −0.7 (τ^2), +5.6 (WB)	No universal "Bangla is worse" claim holds	Gaps reflect evaluation components more than agents
Checker language drives OB gemma gap	15 of 42 PF cases are correct answers in Bangla script; crediting them: +8.0 → +3.0	Benchmark artefact	—
Simulator drives τ^2 gap	35 of 66 PF cases C2; premature-stop flags 46 vs 26	Benchmark artefact; same model plays both roles	A stronger simulator removes most PF cases
Genuine Bangla mechanisms are few	24 of 163 cases C1; name case in both models	Grounding, not tool syntax	Protecting names removes them
gpt-oss gains from Bangla prompt	Invalid actions 1,671 → 955; built-in calls 643 → 99	Format effect	Script switch suppresses native tool-call habit
Failures are mostly language-independent	113 OB subtasks fail in all four runs; 82/112 WB failures C6	Fix the agent loop and benchmarks first	—
Noise is large	TAC reruns change 3 of 13 scores; gap +13.0 vs +17.4 by rerun choice	Single runs cannot attribute small effects	—

Table 5.1. Separation of observed facts, interpretations and hypotheses.

5.2 Recommended improvements

Benchmark and evaluation

- **Language-agnostic checking** (R1): normalise Bengali digits (Bengali digits → ASCII 0–9), accept Bangla answers via an answer-key synonym list or a semantic judge, or state the answer language in the task. Apply the same rule to TAC keyword checkpoints and τ² communicate checks.
- **Repair benchmark defects** (R2): audit the 57 OfficeBench subtasks where a defect decides a run (e.g., 1-2/0 requires a “meeting” event the task never asks for) and document tool semantics (write_to_file appends).
- **Constrain the user simulator** (R3): reject a user turn that confirms and stops at once; require device-tool calls for device actions; check that the simulator speaks the target language; consider a fixed, stronger simulator model independent of the agent.
- **Case-insensitive identity lookups** (R5) in mailbox and calendar tools, or canonical Latin names in prompts.

Translation

- **Protected tokens** (R4, R5): keep person names, file names, quoted literals and grader strings in Latin script, use ASCII digits, and verify automatically. For OfficeBench this is implemented in the v2 translation (298/298 tasks pass the automatic checks; 294 strings changed; Bob kept in 54/54 tasks).
- **Native-speaker review** of relative-date phrases and domain terms (R7) before claiming genuine effects.

Agent and harness

- **Verification before finishing** (R12): treat an empty result as a signal to retry (case variants, alternative tool, filesystem search) rather than as the answer; re-read the changed state before claiming success.
- **Stuck recovery** (R8, R9): after two identical rejected actions, force a strategy change (chunked or append writes, shorter text); expose the reply cap and the append semantics in the tool description; raise the 512-token cap for long writes.
- **Format robustness** (R8): use native function calling or constrained decoding for gpt-oss; return a structured error naming the invalid argument instead of crashing the run (WorkBench invented argument names).
- **Fix harness limits** (R15): raise the llm app limit above 50 tokens for reasoning models; retry empty model replies; quote shell arguments; pre-authorise ownCloud sessions where the task does not test authentication.
- **Completion checklists** (R13, R14): derive sub-goals from the task and check them before finish_task; reserve budget for three-app tasks.

5.3 Follow-up experiments

#	Experiment	Tests	Prediction if the hypothesis holds	Metric
E1	Rerun OfficeBench Bangla with the v2 translation (names Latin, ASCII digits, literals protected). A rerun was started on 2026-10-10 (tag bn2) and has not completed; its results are not part of this report.	R4, R5, R6	NC and TX cases disappear; gemma gap shrinks toward the S9 value	PF count by code; gap
E2	Answer-language control: “answer in English” instruction vs multilingual checker	R1	Both remove the 15 AL cases; gap ≈ +3 pp	Gap, AL count
E3	≥ 3 seeds per cell on all benchmarks	R17	Discordant counts shrink; CIs reflect variance	Per-task pass rate, ICC
E4	τ²: Bangla agent with a fixed strong simulator; and English simulator with Bangla agent	R3	Most C2 cases pass; residual gap ≈ C1 + C6	Gap, premature-stop rate
E5	gpt-oss format ablation: native tool calling vs text JSON, in both languages	R8 (H)	Bangla advantage disappears under native tool calling	Invalid actions, gap
E6	Verification ablation: auto-retry empty lookups with case variants; chunked-write recovery	R5, R9, R12	C1 name-case and TL cases pass	PF count; steps used
E7	Reply-cap ablation (512 → 2048 tokens) and documented append semantics	R9	Cut-off loops vanish in both languages	TL count
E8	Relative-date probe set reviewed by native speakers	R7	Rolling-window grounding persists only if genuine	Accuracy on probes

#	Experiment	Tests	Prediction if the hypothesis holds	Metric
E9	Symmetric audit of EN-fail/BN-pass cases (OfficeBench, τ²) and native-speaker second review	Bias in S-scenarios	Symmetric gaps match S9/S7	κ, gap
E10	Scale TAC beyond 23 task-runs with English reruns too	Power	Gap estimate stabilises	Gap CI width

Table 5.2. Proposed experiments, each tied to a failure mechanism.

5.4 Limitations of this synthesis

- One run per cell at temperature 0 in every benchmark (only some TAC Bangla tasks were rerun).
- All annotators are LLMs; second review exists for WorkBench and part of τ²; agreement on fine confounds is modest (τ² κ = 0.39).
- Different task sets and sample sizes (23 to 300 pairs); TAC and WorkBench are pilots.
- Heuristic signals: the τ² premature-stop detector has about 50% precision; gpt-oss OfficeBench Bangla failures beyond the first 51 subtasks are not root-cause coded.
- Bangla tasks were machine-translated by the same model family (gemma4:31b); translation noise is part of the Bangla condition.
- This document transcribes the source reports; it did not recompute any statistic.

Appendix A. Glossary of codes

Code	Meaning
PP / PF / FP / FF	Both pass / English pass, Bangla fail / English fail, Bangla pass / both fail
MELR	Multilingual expected loss ratio = (EN − BN) / EN
BM	Benchmark defect (check asks for what the task does not; wrong reference; app corrupts output)
WC / TR	Wrong content / task read differently
SB / TL / DL	Out of steps / cut-off output loop at the reply cap / reasoning loop
FMT / ER / PF / CAP	Action-format failure / empty result, gave up / premature finish / capability claim or refusal
AL / NC / WD / TX	Answer language / name case / word choice / translation shift (Bangla cells only)
OTH / INF	Other / infrastructure (crashed runs, empty API replies)
C-AMBIG, C-WEAKCHECK, C-HARNESS, C-LLMAPP	Contributing tags: ambiguous task, weak check, harness issue, llm-app issue
S0–S9	Sensitivity scenarios of the source reports (S0 = as scored)

Appendix B. Source-report cross-reference

Topic	OfficeBench report	τ²-bench report	WorkBench report	TAC report
Headline results	R, §2, §11	R, §2, §11	R, §2, §11	R, §2, §11
Trace notes	App. A (gemma), App. D (gpt-oss)	App. A (66 cases)	App. A, B, C	App. A (8 cases)
Sensitivity	§11 (S0–S9), II.10	§11 (S0–S7)	§11 (S0–S6)	§11 (S0–S5)
Case studies	§15 (3-68/0, 1-10/0, 1-15/0)	§15 (airline 19, 9, 8)	§15 (email:41, crm:77, analytics:98)	§15 (T01, T03, T08)